

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		all three needed chloride – white bromide – cream (accept off-white) yellow – iodide	1			1		1
		(ii)		$\Delta_f H^\theta (\text{AgBr}) = 105.6 + (\frac{1}{2} \times -243.1) - 84.4 \quad (1)$ $\Delta_f H^\theta (\text{AgBr}) = -100.35 \quad (1)$ AgCl more stable than AgBr (ecf possible from values) (1) as standard enthalpy of formation is more negative (1)		2	2	4	1	
	(b)	(i)	I	$\text{Cu} + 2\text{Ag}^+ \rightarrow \text{Cu}^{2+} + 2\text{Ag}$		1		1		
			II	$E^\theta (\text{Cu})$ must be more negative/less positive than +0.80V (1) award (1) for either of following reasons reaction is feasible when EMF is positive reaction is feasible when reduction half-cell is more positive than oxidation half-cell			2	2		